

26021 Kemi (Polyteknisk grundlag)/26021 Chemistry (Polytechnical Foundation) reexam August 2025 non random version

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

Der er altid netop ét svar som er mere rigtigt end de andre

Studerende kan kun vælge ét svar per spørgsmål

Hvert rigtigt svar giver 1 point

Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

”,” bruges som decimalseparator i de danske spørgsmål og i svarmulighederne mens ”.” bruges som

decimalseparator i de engelske spørgsmål.

The following approach to scoring responses is implemented and is based on "One best answer"

There is always only one correct answer – a response that is more correct than the rest

Students are only able to select one answer per question

Every correct answer corresponds to 1 point

Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

”,” is used as decimal separator in the Danish questions and in the answer options, while ”.” is used as decimal separator in the English questions.

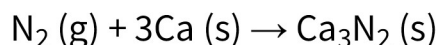
Lithiumjernfosfat, LiFePO_4 , bruges som katodemateriale i visse typer lithium-ion-batterier. Hvad er masseprocenten af lithium i LiFePO_4 ?

Lithium iron phosphate, LiFePO_4 , is used as a cathode material in certain types of lithium-ion batteries. What is the mass percentage of lithium in LiFePO_4 ?

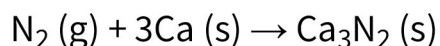
Choose one answer

- ☐ 22,7%
- ☐ 35,4 %
- ☐ 4,4%
- ☐ 6,9%
- ☐ 5,8%

Kvælstof, N_2 , reagerer med calcium (i overskud) og danner calciumnitrid, Ca_3N_2 , ifølge reaktionen nedenfor. Hvad er det minimale volumen kvælstof (ideel gas ved 25 °C og 1 atm), der kræves for at danne 5,0 g calciumnitrid?



Nitrogen gas, N_2 , reacts with excess calcium to form calcium nitride, Ca_3N_2 , according to the reaction below. What is the minimum volume of nitrogen gas (ideal gas at 25 °C and 1 atm) required to form 5.0 g of calcium nitride?



Choose one answer

- ☐ 1,65 L
- ☐ 0,83 L
- ☐ 29,7 L
- ☐ 2,48 L
- ☐ 0,41 L

Et molekyle består af en aromatisk ring med følgende funktionelle grupper bundet til sig:

- 1) En forgrenet alkylkæde
- 2) En alkylkæde indeholdende en amidbinding

Hvilken af følgende strukturer passer til denne beskrivelse?

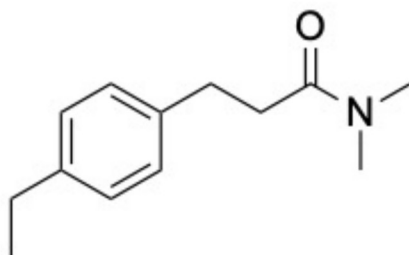
A molecule consists of an aromatic ring with the following functional groups attached:

- 1) A branched alkyl chain
- 2) An alkyl chain containing an amide bond

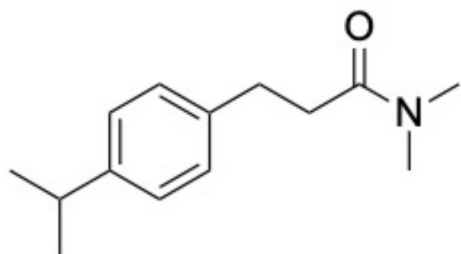
Which of the following structures fits with this description?

Choose one answer

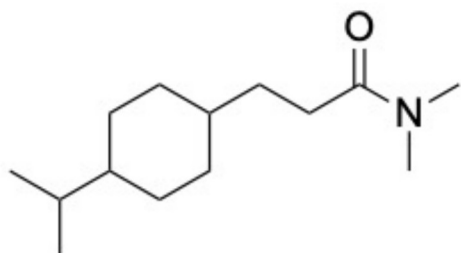
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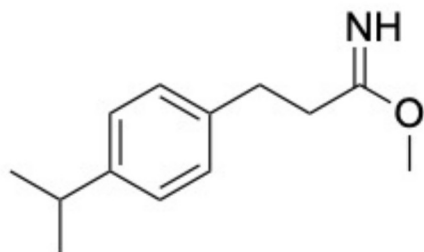
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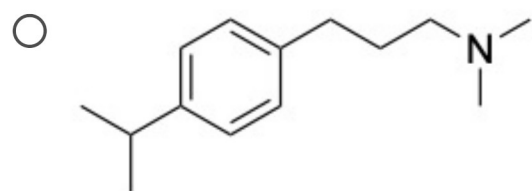


☐



☐





Identificér den ion, der ikke er isoelektronisk med neon.

Identify the ion that is not isoelectronic with neon.

Choose one answer

☐ Na^+

☐ S^{2-}

☐ O^{2-}

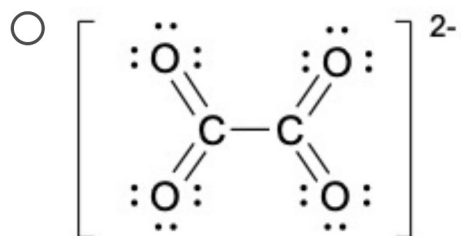
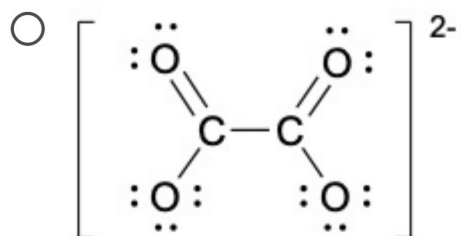
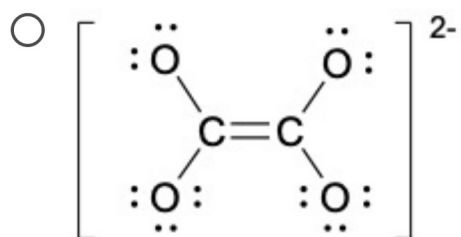
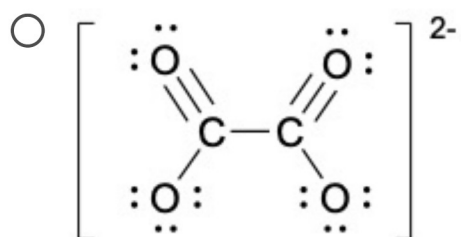
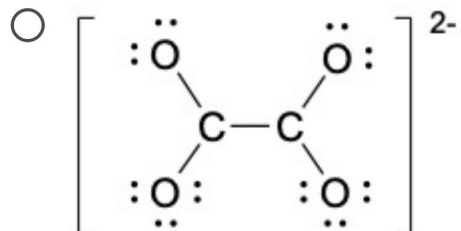
☐ F^-

☐ Al^{3+}

Hvilken af følgende illustrationer repræsenterer en gyldig Lewis-struktur for oxalatanionen ($\text{C}_2\text{O}_4^{2-}$)?

Which of the following illustrations represents a valid Lewis structure for the oxalate anion ($\text{C}_2\text{O}_4^{2-}$)?

Choose one answer



Arranger følgende diatomiske forbindelser i rækkefølge efter stigende kovalent bindingskarakter (dvs. mest kovalent til højre og mindst kovalent til venstre):

HI, NaBr, CsBr, Cl₂, LiF

Arrange the following diatomic compounds in order of increasing covalent character of the bond (i.e. most covalent to the right and least covalent to the left):

HI, NaBr, CsBr, Cl₂, LiF

Choose one answer

- ☐ LiF < CsBr < NaBr < HI < Cl₂
- ☐ NaBr < HI < Cl₂ < LiF < CsBr
- ☐ Cl₂ < LiF < CsBr < NaBr < HI
- ☐ HI < Cl₂ < LiF < CsBr < NaBr
- ☐ CsBr < NaBr < HI < Cl₂ < LiF

Til en vandig NaCl-opløsning med et volumen på 250 mL blev overskydende vandig AgNO_3 tilsat. Det dannede bundfald blev isoleret ved filtrering og vejet: 13,5 g. Bestem den molære koncentration af den vandige NaCl-opløsning før tilsætning af AgNO_3 .

To an aqueous solution of NaCl with a volume of 250 mL, excess aqueous AgNO_3 was added. The formed precipitate was isolated by filtration and weighed: 13.5 g. Determine the molar concentration of the aqueous NaCl solution before AgNO_3 addition.

Choose one answer

- ☐ 0,92 M
- ☐ 0,05 M
- ☐ 2,65 M
- ☐ 0,13 M
- ☐ 0,38 M

Termisk nedbrydning af kalksten, som hovedsageligt består af CaCO_3 , er et nøgletrin i cementproduktion og sker gennem følgende reaktion:



Den globale produktion af CaO estimeres til omkring 283×10^9 kg per år. Biproduktet, CO_2 , frigives til luften. Hvor mange træer ville være nødvendige for at absorbere den CO_2 , der frigives fra denne proces, givet at et gennemsnitligt træ kan binde 21 kg CO_2 fra luften per år?

The thermal decomposition of limestone, which is composed mostly of CaCO_3 , is a key step in cement production and occurs through the following reaction:

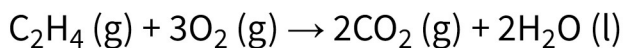


The global production of CaO is estimated to around 283×10^9 kg per year. The byproduct, CO_2 , is released into the air. How many trees would be needed to absorb the CO_2 that is released from this process, given that an average tree can bind 21 kg of CO_2 from the air per year?

Choose one answer

- ☐ $1,1 \times 10^7$ træer (trees)
- ☐ $1,1 \times 10^{10}$ træer (trees)
- ☐ $1,3 \times 10^{12}$ træer (trees)
- ☐ $9,8 \times 10^5$ træer (trees)
- ☐ $2,9 \times 10^3$ træer (trees)

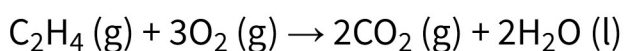
Ethen brænder i ilt ifølge reaktionen nedenfor.



Beregn mængden af varme, der frigives, når 10,0 L C_2H_4 (idealgas ved 25 °C og 1 atm) brændes i overskud af ilt ved konstant tryk, givet at:

	$\text{C}_2\text{H}_4(\text{g})$	$\text{O}_2(\text{g})$	$\text{CO}_2(\text{g})$	$\text{H}_2\text{O}(\text{l})$
ΔH_f° (kJ/mol)	52.4	0	-393.5	-285.8

Ethene burns in oxygen according to the reaction below.



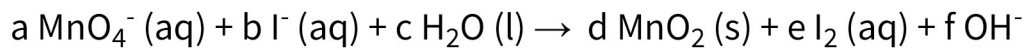
Calculate the amount of heat released when 10.0 L of C_2H_4 (ideal gas at 25 °C and 1 atm) is burned in excess oxygen at constant pressure, given that:

	$\text{C}_2\text{H}_4(\text{g})$	$\text{O}_2(\text{g})$	$\text{CO}_2(\text{g})$	$\text{H}_2\text{O}(\text{l})$
ΔH_f° (kJ/mol)	52.4	0	-393.5	-285.8

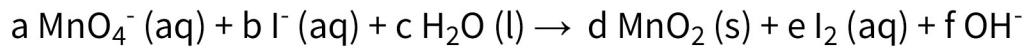
Choose one answer

- ☐ 576 kJ
- ☐ 1411 kJ
- ☐ 97 kJ
- ☐ 338 kJ
- ☐ 201 kJ

Angiv hvilket sæt af støkiometriske koefficienter (a-f) der balancerer redoxreaktionen nedenfor i et vandigt alkalisk miljø:



Indicate which set of stoichiometric coefficients (a-f) that balances the redox reaction below in aqueous alkaline environment:



Choose one answer

- ☐ a = 1, b = 5, c = 8, d = 1, e = 3, f = 8
- ☐ a = 4, b = 10, c = 13, d = 4, e = 5, f = 26
- ☐ a = 2, b = 10, c = 10, d = 2, e = 5, f = 20
- ☐ a = 2, b = 6, c = 4, d = 2, e = 3, f = 8
- ☐ a = 6, b = 10, c = 13, d = 6, e = 5, f = 26

Gibbs' frie energiændring, ΔG^0 , for en redoxreaktion er -356,9 kJ/mol. En elektrokemisk celle konstrueret baseret på denne redoxreaktion bruger følgende halvcellereaktion for katoden:



Identificér den halvcellereaktion, der blev brugt til anoden.

The Gibbs free energy change, ΔG^0 , for a redox reaction is -356.9 kJ/mol. An electrochemical cell constructed based on this redox reaction uses the following reduction half-reaction for the cathode:



Identify the half reaction that was used for the anode.

Choose one answer

- ☐ $\text{Zn (s)} \rightarrow \text{Zn}^{2+} (\text{aq}) + 2\text{e}^- \quad E_{\text{ox}}^0 = 0,762 \text{ V}$
- ☐ $\text{Pt (s)} \rightarrow \text{Pt}^{2+} (\text{aq}) + 2\text{e}^- \quad E_{\text{ox}}^0 = -1,180 \text{ V}$
- ☐ $\text{Pb (s)} \rightarrow \text{Pb}^{2+} (\text{aq}) + 2\text{e}^- \quad E_{\text{ox}}^0 = 0,126 \text{ V}$
- ☐ $\text{Ca (s)} \rightarrow \text{Ca}^{2+} (\text{aq}) + 2\text{e}^- \quad E_{\text{ox}}^0 = 2,868 \text{ V}$
- ☐ $\text{Ni (s)} \rightarrow \text{Ni}^{2+} (\text{aq}) + 2\text{e}^- \quad E_{\text{ox}}^0 = 0,257 \text{ V}$

Hvilket af følgende grundstoffer kræver den højeste energi for at fjerne en elektron i gasfasen?

Which of the following atoms has the highest energy requirement for removal of an electron in the gas phase?

Choose one answer

☐ Se

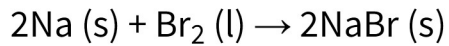
☐ O

☐ K

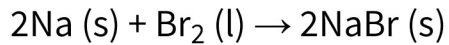
☐ Fe

☐ Li

2,30 g natrium (Na) og 9,59 g brom (Br₂) reagerer og danner natriumbromid (NaBr) ifølge reaktionen nedenfor. 7,85 g natriumbromid blev dannet. Hvad var udbyttet af natriumbromid i procent?



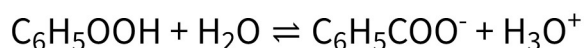
2.30 g of sodium (Na) and 9.59 g of bromine (Br₂) react to form sodium bromide (NaBr) according to the equation below. After the reaction, 7.85 g of sodium bromide was obtained. What was the percent yield of sodium bromide?



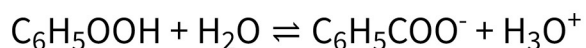
Choose one answer

- ☐ 53%
- ☐ 76%
- ☐ 66%
- ☐ 93%
- ☐ 32%

Benzoesyre opløses i vand i en koncentration på 0,150 M, og den nedenfor viste ligevægt etableres. Hvad er opløsningens pH ved 25 °C, givet at pK_a -værdien for benzoesyre i vand er 4,20?



Benzoic acid is dissolved in water at concentration of 0.150 M, and the equilibrium shown below is established. What is the pH of the solution at 25 °C, given that the pK_a value of benzoic acid is 4.20 in water?



Choose one answer

- ☐ 2,5
- ☐ 2,9
- ☐ 3,3
- ☐ 3,7
- ☐ 4,8

Dimethylsulfoxid (DMSO) er et organisk opløsningsmiddel, der er fuldt blandbart med vand. Beregn frysepunktet for en vandig opløsning af DMSO med en koncentration på 0,60 mol/kg, givet at van't Hoff-faktoren er 1,0, og at $K_f = 3,85\text{ }^{\circ}\text{C/m}$.

Dimethylsulfoxide (DMSO) is an organic solvent that is fully miscible with water. Calculate the freezing point of an aqueous solution of DMSO with a concentration of 0.60 mol/kg, given that the van 't Hoff factor is 1.0 and that $K_f = 3.85\text{ }^{\circ}\text{C/m}$.

Choose one answer

- ☐ -1,41 $^{\circ}\text{C}$
- ☐ -2,31 $^{\circ}\text{C}$
- ☐ -1,21 $^{\circ}\text{C}$
- ☐ 1,21 $^{\circ}\text{C}$
- ☐ 0,0 $^{\circ}\text{C}$

Tantal (Ta) bruges i belægninger til korrosionsbeskyttelse. Det krystalliserer i en rumcentreret kubisk (bcc) struktur og har en densitet på $16,68 \text{ g/cm}^3$. Hvor mange tantalkrystal-enhedsceller er nødvendige for at opnå en $10,0 \text{ }\mu\text{m}$ tyk tantalbelægning på et område på $10,0 \text{ cm}^2$?

Tantalum (Ta) is used in coatings for corrosion protection. It crystallizes in a body-centered cubic (bcc) structure and has a density of 16.68 g/cm^3 . How many tantalum crystal unit cells are needed to obtain a 10.0 mm thick tantalum coating on an area of 10.0 cm^2 ?

Choose one answer

- ☐ $9,2 \times 10^{-4}$
- ☐ $5,3 \times 10^{35}$
- ☐ $2,8 \times 10^{20}$
- ☐ $1,2 \times 10^{10}$
- ☐ $7,6 \times 10^{16}$

Hastighedskonstanten for en reaktion blev fundet til at være $3,0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$ ved 325°C og $20 \text{ M}^{-1} \text{ s}^{-1}$ ved 500°C . Hvad er aktiveringsenergien for denne reaktion?

The rate constant for a reaction was found to be $3.0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$ at 325°C and $20 \text{ M}^{-1} \text{ s}^{-1}$ at 500°C . What is the activation energy for this reaction?

Choose one answer

- ☐ 19 kJ/mol
- ☐ 5 kJ/mol
- ☐ 193 kJ/mol
- ☐ 134 kJ/mol
- ☐ 97 kJ/mol

En mættet vandig opløsning af $\text{Mg}(\text{OH})_2$ fremstilles. Hvad er opløsningens pH-værdi ved 25 °C, givet at opløselighedsproduktet K_{sp} af $\text{Mg}(\text{OH})_2$ i vand ved 25 °C er $5,61 \times 10^{-12}$?

A saturated aqueous solution of $\text{Mg}(\text{OH})_2$ is prepared. What is the pH of the solution at 25 °C, given that the solubility product K_{sp} of $\text{Mg}(\text{OH})_2$ in water at 25 °C is 5.61×10^{-12} ?

Choose one answer

- ☐ 10,4
- ☐ 8,9
- ☐ 13,5
- ☐ 11,3
- ☐ 6,2

En af de stærke linjer i emissionsspektret for helium er ved 501 nm. Hvilken energi repræsenterer denne emissionslinje?

One of the strong lines in the Helium emission spectrum occurs at 501 nm. Which energy does this line represent?

Choose one answer

- ☐ 400 kJ/mol
- ☐ 98 kJ/mol
- ☐ 38 kJ/mol
- ☐ 239 kJ/mol
- ☐ 678 kJ/mol

En pufferopløsning fremstilles og indeholder 1,50 M eddikesyre og 1,50 M natriumacetat. Syredissociationskonstanten K_a for eddikesyre er $1,77 \times 10^{-5}$. Hvad er pH-værdien af denne pufferopløsning?

A buffer solution is prepared and contains 1.50 M acetic acid and 1.50 M sodium acetate. The acid dissociation constant K_a of acetic acid is 1.77×10^{-5} . What is the pH of this buffer solution?

Choose one answer

- ☐ 5,01
- ☐ 4,75
- ☐ 5,75
- ☐ 7,00
- ☐ 6,34

